

HOTEL MANAGEMENT SYSTEM

Desktop Application (Local – Windows PC)

Project Overview

- Manages hotel operations digitally
- Reduces manual work
- Stores data securely in a database

Technologies Used

- Programming Language: C#
- Framework: Windows Forms
- Database: MySQL
- Database Connectivity: `MySql.Data.MySqlClient`
- Configuration Management: Configuration Manager

Software Development Model

- The Waterfall Model is followed
- Development is done in clear, sequential phases
- Each phase is completed before moving to the next

Requirements Gathering

- Requirements were gathered using interviews
- Hotel staff were consulted to understand system needs
- Focus was on daily operations and existing problems

Functional Requirements

- **Add Guest:** enter guest details, validate inputs, store data, show confirmation
- **View All Guests:** retrieve and display records, auto-refresh after changes
- **Search Guest:** search by name or room number, show results or no-match message
- **Check Out Guest:** select guest, confirm deletion, update list
- **Data Validation:** prevent empty fields, invalid dates, non-numeric room numbers
- **Database Connection:** connect at startup, handle connection errors clearly

Functional Requirements

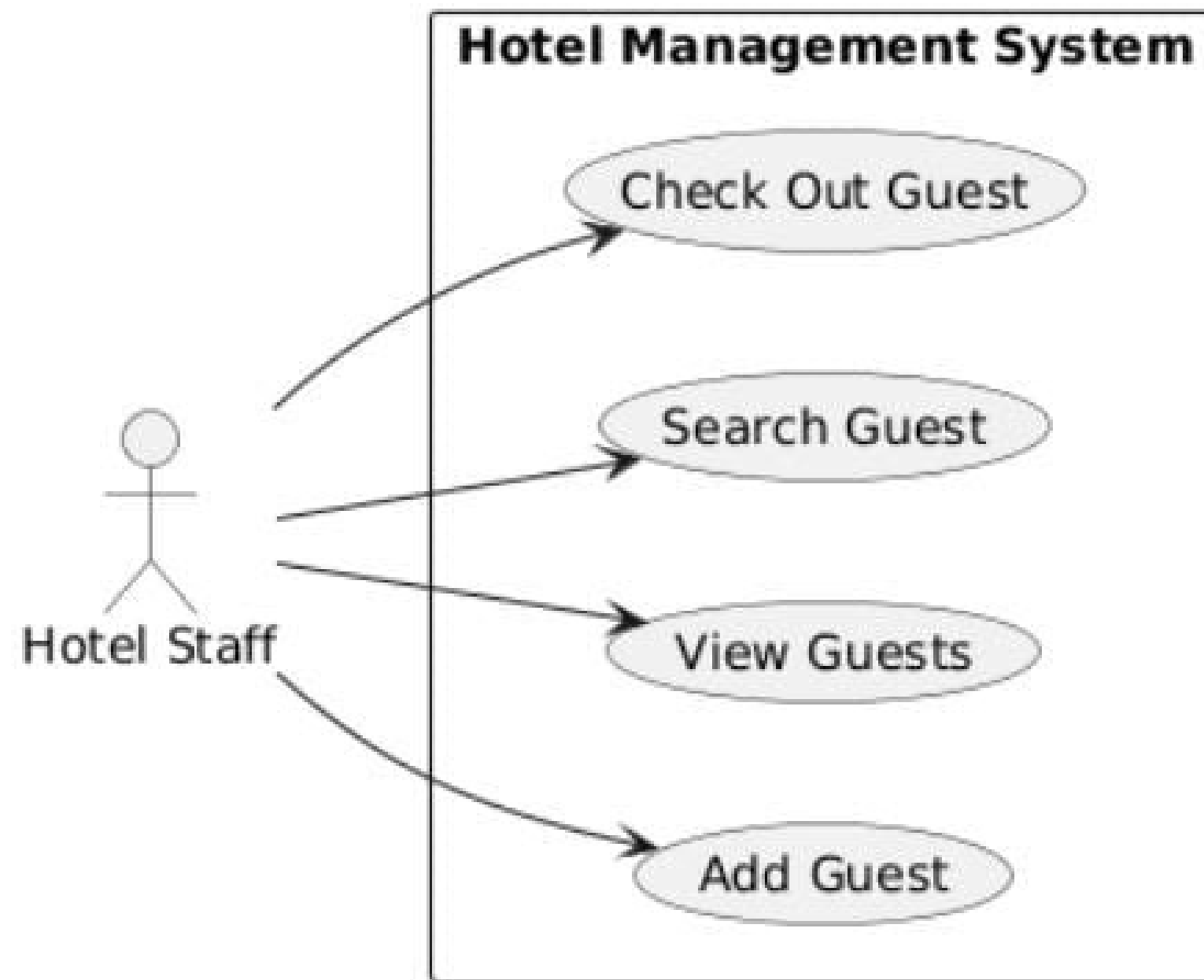
FR1. Add Guest

- 1.1 The system shall allow the staff to enter: Guest Name, Room Number, Check-In Date
- 1.2 The system shall validate that: All fields are filled, Room number is a valid integer.
- 1.3 The system shall insert the new guest record into the Guests table in MySQL.
- 1.4 The system shall display a confirmation message when the record is successfully added

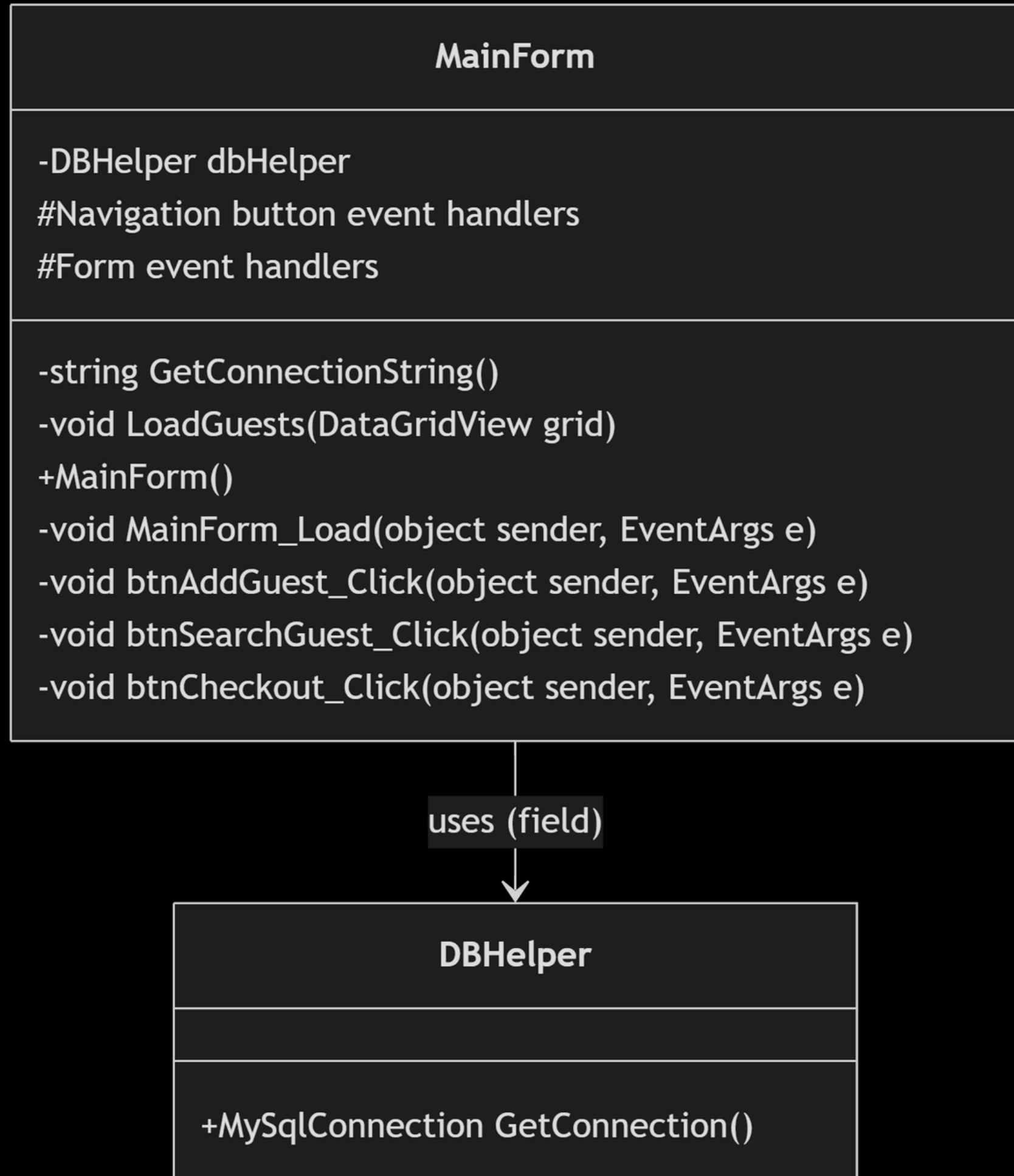
Non-Functional Requirements

- **Performance:** Fast loading and quick database operations
- **Usability:** Simple, clear, and consistent interface
- **Reliability:** Stable database connection, no duplicate IDs
- **Maintainability:** Well-commented code
- **Portability:** Runs on Windows with required dependencies

Use Case Diagram



Class Diagram



System Architecture

- Monolithic structure
- User Interface, business logic, and database access handled in the same form
- Uses code-behind approach
- Direct interaction with MySQL from UI events

Why Monolithic?

- The project is small in scope and simple in functionality
- All features are closely related and managed in one application
- Faster to develop and easier to understand at this stage

Database Design

```
CREATE TABLE Guests (  
  GuestID INT AUTO_INCREMENT PRIMARY KEY,  
  Email VARCHAR(150) DEFAULT "",  
  Name VARCHAR(100) NOT NULL,  
  RoomNumber INT NOT NULL,  
  CheckInDate DATE NOT NULL  
);
```

	GuestID	Name	Email	RoomNumber	CheckInDate
	NULL	NULL	NULL	NULL	NULL

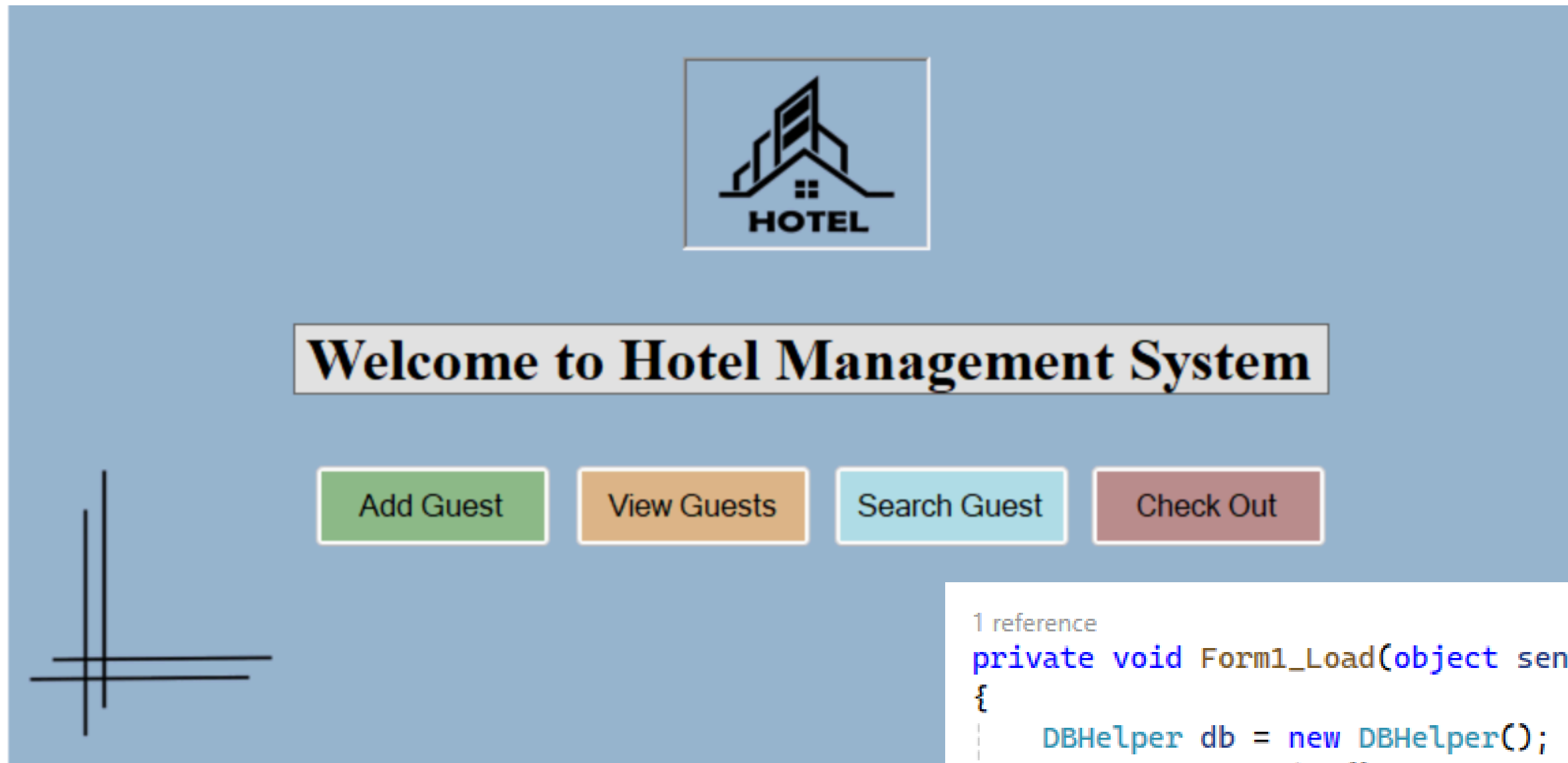
DBHelper Class

```
using MySql.Data.MySqlClient;
using System;
using System.Configuration;
using System.Windows.Forms;

4 references
public class DBHelper
{
    1 reference
    public MySqlConnection GetConnection()
    {
        string connStr = ConfigurationManager.ConnectionStrings["HotelDB"].ConnectionString;
        return new MySqlConnection(connStr);
    }
}
```

- GetConnection()
- Reads connection string from config file
- Returns a MySQL connection

Index Page



1 reference

```
private void Form1_Load(object sender, EventArgs e)
{
    DBHelper db = new DBHelper();
    db.TestConnection();

    LoadGuests(dgvGuests);

    // =====
    // RUN TESTS (TEMPORARY)
    // =====
    RunAllTests();
}
```


Add Guest

Guest Name:

Email:

Room Number:

Check-In Date:

Add Guest

Back



Add Guest

Add Guest

Guest Name:

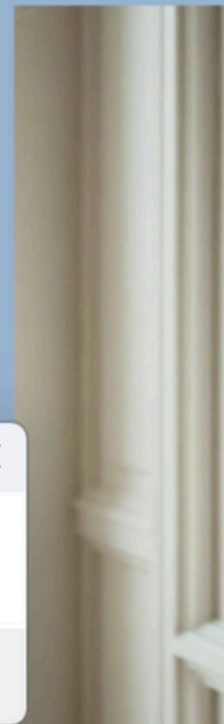
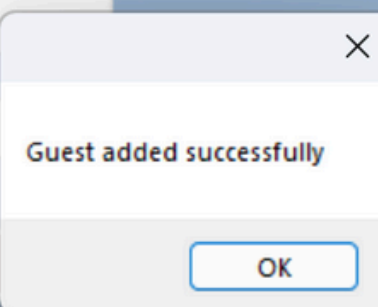
Email:

Room Number:

Check-In Date:

Add Guest

Back



Add Guest

Guest Name:

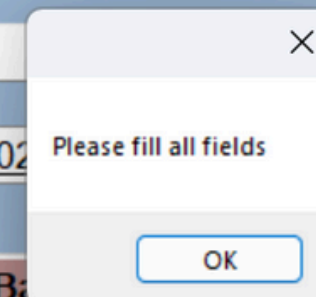
Email:

Room Number:

Check-In Date:

Add Guest

Back



Add Guest

1 reference

```
private void btnAddGuest_Click(object sender, EventArgs e)
{
    if (string.IsNullOrEmpty(txtName.Text) ||
        string.IsNullOrEmpty(txtEmail.Text) ||
        string.IsNullOrEmpty(txtRoomNumber.Text))
    {
        MessageBox.Show("Please fill all fields");
        return;
    }

    if (!int.TryParse(txtRoomNumber.Text, out int roomNumber))
    {
        MessageBox.Show("Room number must be numeric");
        return;
    }

    using (MySqlConnection conn =
        new MySqlConnection(GetConnectionString()))
    {
        conn.Open();

        string query =
            "INSERT INTO Guests (Name, Email, RoomNumber, CheckInDate) " +
            "VALUES (@name, @email, @room, @date)";

        MySqlCommand cmd = new MySqlCommand(query, conn);
        cmd.Parameters.AddWithValue("@name", txtName.Text);
        cmd.Parameters.AddWithValue("@email", txtEmail.Text);
        cmd.Parameters.AddWithValue("@room", roomNumber);
        cmd.Parameters.AddWithValue("@date", dtpCheckIn.Value);

        cmd.ExecuteNonQuery();
    }
}
```

```
{
    conn.Open();

    string query =
        "INSERT INTO Guests (Name, Email, RoomNumber, CheckInDate) " +
        "VALUES (@name, @email, @room, @date)";

    MySqlCommand cmd = new MySqlCommand(query, conn);
    cmd.Parameters.AddWithValue("@name", txtName.Text);
    cmd.Parameters.AddWithValue("@email", txtEmail.Text);
    cmd.Parameters.AddWithValue("@room", roomNumber);
    cmd.Parameters.AddWithValue("@date", dtpCheckIn.Value);

    cmd.ExecuteNonQuery();
}

MessageBox.Show("Guest added successfully");

LoadGuests(dgvGuests);

txtName.Clear();
txtRoomNumber.Clear();
txtEmail.Clear();
}
```

View Guests

View Guests

	GuestId	Name	Email	RoomNumber	CheckInDate
▶	12	John Smith	john.smith@gma...	201	01/02/2025
	13	Emily Brown	emily.brown@ya...	202	02/02/2025
	14	Michael Lee	michael.lee@outl...	203	03/02/2025
	15	Sarah Johnson	sarah.johnson@g...	204	04/02/2025
	16	David Wilson	david.wilson@ho...	205	05/02/2025
	17	Jessica Taylor	jessica.taylor@g...	206	06/02/2025
	18	Daniel Anderson	daniel.anderson...	207	07/02/2025
	19	Laura Martinez	laura.martinez@...	208	08/02/2025
	20	Robert Thomas	robert.thomas@...	209	09/02/2025

Back

View Guests

View

	GuestId	Name	Email
▶	12	John Smith	john.smith@gmail.com
	13	Emily Brown	emily.brown@gmail.com
	14	Michael Lee	michael.lee@gmail.com
	15	Sarah Johnson	sarah.johnson@gmail.com
	16	David Wilson	david.wilson@gmail.com
	17	Jessica Taylor	jessica.taylor@gmail.com
	18	Daniel Anderson	daniel.anderson@gmail.com
	19	Laura Martinez	laura.martinez@gmail.com
	20	Robert Thomas	robert.thomas@gmail.com

```
private void LoadGuests(DataGridView grid)
{
    using (MySqlConnection conn =
        new MySqlConnection(GetConnectionString()))
    {
        conn.Open();

        string query =
            "SELECT GuestId, Name, Email, RoomNumber, CheckInDate FROM Guests";

        MySqlDataAdapter da = new MySqlDataAdapter(query, conn);
        DataTable dt = new DataTable();
        da.Fill(dt);

        grid.DataSource = dt;
        grid.AutoSizeColumnsMode = DataGridViewAutoSizeColumnsMode.Fill;
        grid.ClearSelection();

        grid.ReadOnly = true;
        grid.SelectionMode = DataGridViewSelectionMode.FullRowSelect;
        grid.MultiSelect = false;

        grid.RowsDefaultCellStyle.Font =
            new Font("Segoe UI", 11);
        grid.AlternatingRowsDefaultCellStyle.Font =
            new Font("Segoe UI", 11);

        grid.EnableHeadersVisualStyles = false;
        grid.ColumnHeadersDefaultCellStyle.BackColor =
            Color.DarkSlateGray;
        grid.ColumnHeadersDefaultCellStyle.ForeColor =
            Color.White;
        grid.ColumnHeadersDefaultCellStyle.Font =
            new Font("Segoe UI", 11, FontStyle.Bold);
    }
}
```

Search Guest

Search Guests

Guest Name:

Room Number:

Search Guest

Back

Search Guest

Search Guest

Guest Name:

1 reference

```
private void btnSearchGuest_Click(object sender, EventArgs e)
{
    string guestName = textBoxNameSearch.Text.Trim();
    string roomNumber = textBoxRoomSearch.Text.Trim();

    using (MySqlConnection con =
        new MySqlConnection(GetConnectionString()))
    {
        string query = "SELECT * FROM Guests WHERE 1=1";

        if (!string.IsNullOrEmpty(guestName))
            query += " AND LOWER(Name) LIKE LOWER(@name)";

        if (!string.IsNullOrEmpty(roomNumber))
            query += " AND RoomNumber = @room";

        MySqlCommand cmd = new MySqlCommand(query, con);

        if (!string.IsNullOrEmpty(guestName))
            cmd.Parameters.AddWithValue("@name",
                "%" + guestName + "%");

        if (!string.IsNullOrEmpty(roomNumber))
            cmd.Parameters.AddWithValue("@room", roomNumber);

        MySqlDataAdapter da = new MySqlDataAdapter(cmd);
        DataTable dt = new DataTable();
        da.Fill(dt);

        dgvSearchGuests.DataSource =
            dt.Rows.Count > 0 ? dt : null;

        if (dt.Rows.Count == 0)
            MessageBox.Show("No guest found.");
    }
}
```

Check Out

Checkout

	GuestId	Name	Email	RoomNumber	CheckInDate
▶	12	John Smith	john.smith@gmai...	201	01/02/2025
	13	Emily Brown	emily.brown@yah...	202	02/02/2025
	14	Michael Lee	michael.lee@...	203	03/02/2025
	15	Sarah Johnson	sarah.johnso...	204	04/02/2025
	16	David Wilson	david.wilson...	205	05/02/2025
	17	Jessica Taylor	jessica.taylo...	206	06/02/2025
	18	Daniel Anderson	daniel.ander...	207	07/02/2025
	19	Laura Martinez	laura.martin...	208	08/02/2025

Checkout

	Name	Email	RoomNumber
	John Smith	john.smith@gmai...	201
	Emily Brown	emily.brown@yah...	202
	Laura Martinez	laura.martinez@o...	208

Checkout Complete

 John Smith has been checked out successfully.

OK

Checkout

Back

Checkout

	GuestId	Name	Email	RoomNumber	CheckInDate
▶	12	John Smith	john.smith@gmai...	201	01/02/2025
	13	Emily Brown	emily.brown@yah...	202	02/02/2025
	14	Michael Lee	michael.lee@...	203	03/02/2025
	15	Sarah Johnson	sarah.johnso...	204	04/02/2025
	16	David Wilson	david.wilson...	205	05/02/2025
	17	Jessica Taylor	jessica.taylo...	206	06/02/2025
	18	Daniel Anderson	daniel.ander...	207	07/02/2025
	19	Laura Martinez	laura.martin...	208	08/02/2025

Confirm Checkout

 Are you sure you want to check out John Smith (Guest ID: 12)?

Yes

No

Checkout

Back

Check Out

1 reference

```
private void btnCheckout_Click(object sender, EventArgs e)
{
    if (dgvCheckout.SelectedRows.Count == 0)
    {
        MessageBox.Show(
            "Please select a guest to check out.",
            "Checkout",
            MessageBoxButtons.OK,
            MessageBoxIcon.Warning);
        return;
    }

    int guestId = Convert.ToInt32(
        dgvCheckout.SelectedRows[0].Cells["GuestID"].Value);

    string guestName =
        dgvCheckout.SelectedRows[0].Cells["Name"].Value.ToString();

    DialogResult confirm = MessageBox.Show(
        $"Are you sure you want to check out {guestName} (Guest ID: {guestId})?",
        "Confirm Checkout",
        MessageBoxButtons.YesNo,
        MessageBoxIcon.Question);

    if (confirm != DialogResult.Yes)
        return;

    using (MySQLConnection con =
        new MySQLConnection(GetConnectionString()))
    {
        con.Open();

        string query =
            "DELETE FROM Guests WHERE GuestID = @guestId";

        MySqlCommand cmd = new MySqlCommand(query, con);
        cmd.Parameters.AddWithValue("@guestId", guestId);
        cmd.ExecuteNonQuery();
    }
}
```

	RoomNumber	CheckInDate
h@mai...	201	01/02/2025
wn@yah...	202	02/02/2025
ee@outl...	203	03/02/2025
nson@g...	204	04/02/2025
son@hot...	205	05/02/2025
ylor@g...	206	06/02/2025
derson...	207	07/02/2025
	208	08/02/2025

```
MessageBox.Show(
    $"{guestName} has been checked out successfully.",
    "Checkout Complete",
    MessageBoxButtons.OK,
    MessageBoxIcon.Information);
```

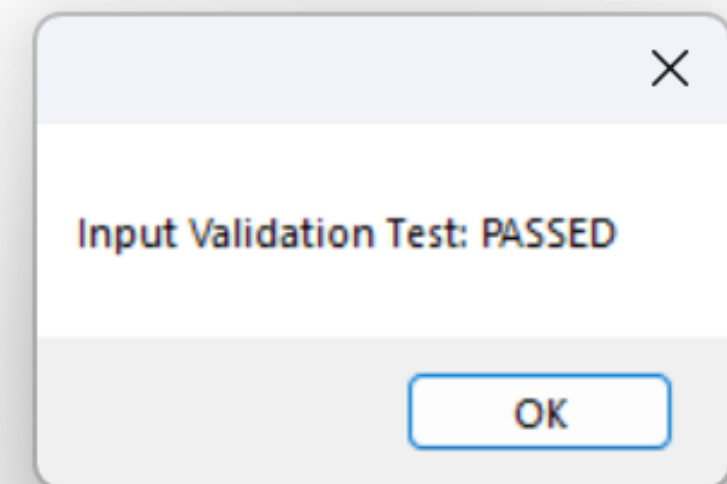
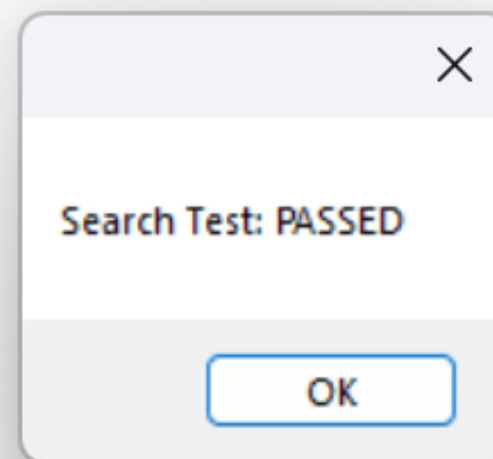
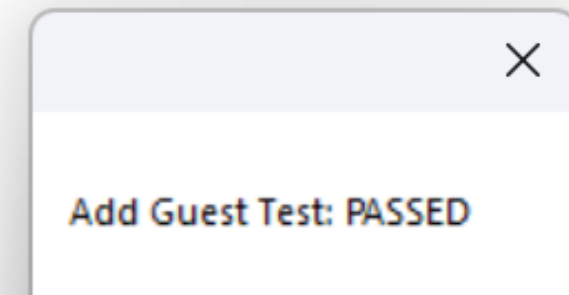
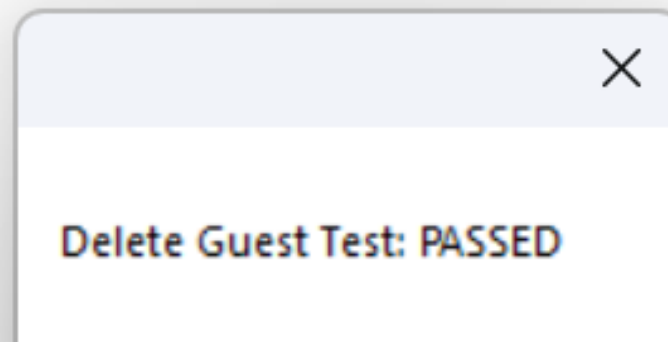
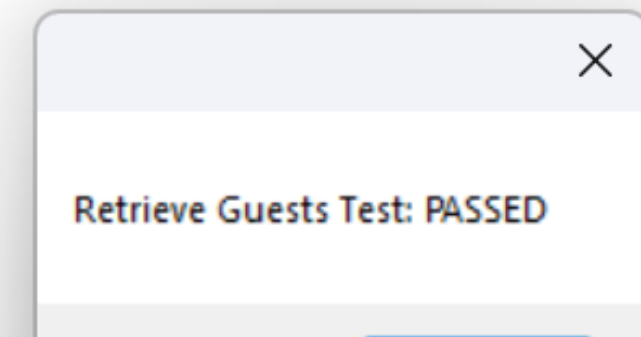
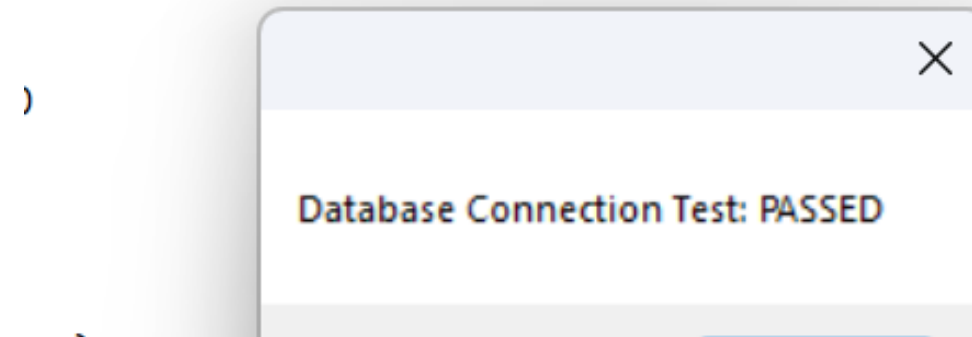
```
dgvCheckout.Rows.RemoveAt(
    dgvCheckout.SelectedRows[0].Index);
```

Testing

```
// =====  
// TESTING METHODS (ADDED)  
// =====
```

1 reference

```
private void RunAllTests()  
{  
    TestDatabaseConnection();  
    TestAddGuest();  
    TestRetrieveGuests();  
    TestSearch();  
    TestDeleteGuest();  
    TestValidation();  
}
```

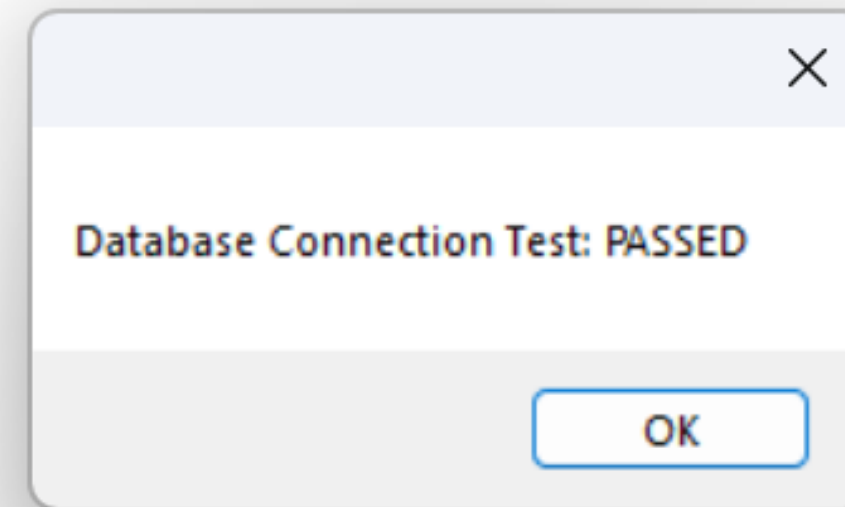
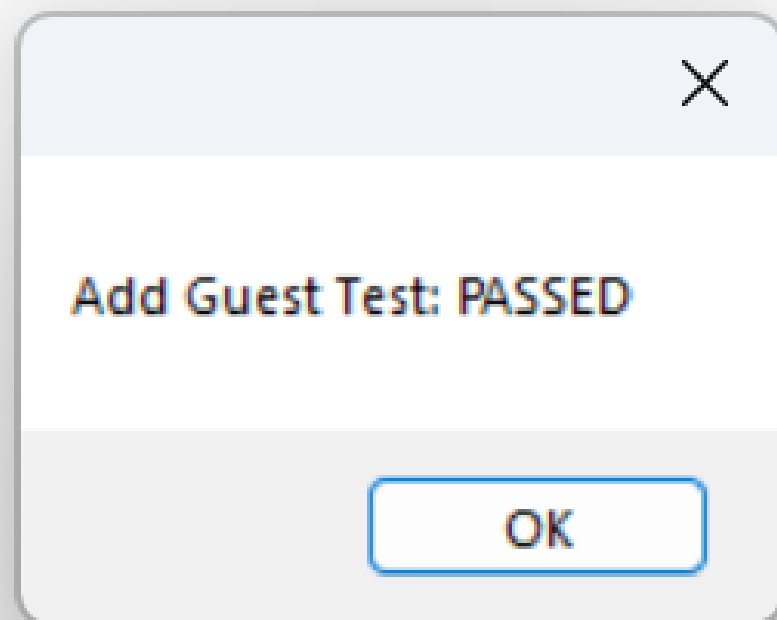


Testing

1 reference

```
private void TestDatabaseConnection()
{
    try
    {
        using (MySqlConnection con =
            new MySqlConnection(GetConnectionString()))
        {
            con.Open();
        }

        MessageBox.Show("Database Connection Test: PASSED");
    }
    catch
    {
        MessageBox.Show("Database Connection Test: FAILED");
    }
}
```



1 reference

```
private void TestAddGuest()
{
    try
    {
        int id = AddTestGuest();
        MessageBox.Show(id > 0
            ? "Add Guest Test: PASSED"
            : "Add Guest Test: FAILED");
    }
    catch
    {
        MessageBox.Show("Add Guest Test: FAILED");
    }
}
```

The slide features a white background with a yellow vertical bar on the left and a dark gray vertical bar on the right. Both bars contain white geometric shapes, including triangles and diamonds. The text "Thank You" is centered in a large, bold, black sans-serif font.

Thank You

For your attention

